

Variable displacement axial piston pump

Series AP · model AP-45

Product data sheet · document 2014-AP-045-EN · issue 2 (07/2015)

General

The AP series covers variable displacement axial piston pumps for open circuit applications in industrial machinery. Model AP-45 is supplied with pressure compensator as standard. Direction of rotation is clockwise when viewed on the shaft end. Counter-clockwise version is not available for this frame size.

Performance data

Description	Value
Displacement	45 cm ³ /rev
Rated pressure	25 MPa
Peak pressure	28 MPa
Maximum speed	2 200 rpm
Minimum speed	600 rpm
Flow at 1 450 rpm	62,1 l/min
Volumetric efficiency	0,93
Overall efficiency	0,86

Fluid and operating conditions

Description	Value
Hydraulic fluid	Mineral oil HLP to DIN 51524-2
Viscosity range	15 – 100 mm ² /s
Fluid temperature	–15 °C to +80 °C
Ambient temperature	not stated
Required fluid cleanliness	ISO 4406 20/18/15
Minimum inlet pressure	0,85 bar abs
Maximum case pressure	1,5 bar abs

Installation

The pump is designed for flange mounting to ISO 3019-2 (metric, 2-bolt). Shaft is splined, 14 tooth, 12/24 DP. Pressure port is SAE 6000 psi 3/4 inch. Suction port is SAE 2500 psi 1 1/2 inch. One drain port G 1/2 inch is provided on the top of the housing.

Mounting position is optional. When mounting with the shaft upwards the case must be filled with fluid before start-up.

Ordering information

Code	Description
AP-45-PC-S14	Pressure compensator, ISO flange, splined shaft 14T
AP-45-LS-S14	Load sensing control, ISO flange, splined shaft 14T

Service

Recommended filter element replacement interval is 1 500 operating hours. Expected service life at rated pressure is 10 000 operating hours under stated cleanliness conditions. Spare parts for series AP are available until 12/2027.

Note: weight, dimensions, noise emission data and radial shaft load limits are given in the AP series installation manual, document 2014-AP-IM-EN, which is not part of this data sheet.

AXIOMA Hydraulics is a fictitious brand created for training purposes. Data in this document are illustrative only.